# A Proof of Graffiti.pc Conjecture 387

An Upper-Median Degree Bound for the 2-Domination Number

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**Abstract**

Let $G$ be a finite simple graph on $n \geq 1$ vertices, and let $m$ be the upper median of its degree sequence. We prove that the 2-domination number of $G$ satisfies

$$\gamma_2(G) \leq n - m + 1.$$

The proof passes to the complement graph, separates the vertices according to a degree threshold, and handles the nontrivial case by associating a family of polynomials to the low-degree vertices of the complement. A minimally linearly dependent subfamily yields a small set whose common zero structure controls the number of complement-neighbours of every remaining vertex. The argument does not use connectedness.

## 1 Definitions and statement

Let $G$ be a finite simple graph with vertex set $V(G)$ and order $|V(G)| = n$. Arrange the degrees of $G$ in nondecreasing order:

$$d_1 \leq d_2 \leq \cdots \leq d_n.$$

The *upper median degree* is

$$m := d_{\lfloor n/2 \rfloor + 1}.$$

**Definition 1.** A set $D \subseteq V(G)$ is *2-dominating* if every vertex in $V(G) \setminus D$ has at least two neighbours in $D$. The minimum cardinality of a 2-dominating set is denoted by $\gamma_2(G)$.

**Theorem 1** (Graffiti.pc Conjecture 387)**.** *For every finite simple graph $G$ on $n \geq 1$ vertices,*

$$\gamma_2(G) \leq n - m + 1,$$

*where $m$ is the upper median degree of $G$.*

## 2 Proof

*Proof.* If $m = 0$, then $D = V(G)$ is 2-dominating and $\gamma_2(G) \leq n \leq n - m + 1$. Henceforth assume $m \geq 1$.

Let

$$F := \overline{G}, \qquad t := n - 1 - m, \qquad q := t + 2 = n - m + 1.$$

Because $m \geq 1$, we have $q \leq n$. For a vertex $x$, write $d_F(x)$ for its degree in $F$.

Partition the vertices into

$$H := \{x \in V(G) : d_F(x) \leq t\}, \qquad B := V(G) \setminus H,$$

and set

$$h := |H|, \qquad k := |B|.$$

By the definition of the upper median, at least $\lceil n/2 \rceil$ vertices have degree at least $m$ in $G$. Since

$$d_F(x) = n - 1 - d_G(x),$$

these vertices have $F$-degree at most $n - 1 - m = t$. Therefore

$$h \geq \left\lceil \frac{n}{2} \right\rceil, \qquad k \leq \left\lfloor \frac{n}{2} \right\rfloor, \qquad h \geq k. \tag{1}$$

**Case 1:** $k \leq t + 1$**.** Choose a set $D \subseteq V(G)$ of cardinality $q = t + 2$ such that $B \subseteq D$. Such a set exists because $k < q \leq n$. Every vertex $v \in V(G) \setminus D$ belongs to $H$, and hence

$$d_F(v, D) \leq d_F(v) \leq t.$$

Thus at most $t$ vertices of $D$ are nonneighbours of $v$ in $G$. Since $|D| = q = t + 2$, the vertex $v$ has at least

$$q - t = 2$$

neighbours in $D$. Hence $D$ is 2-dominating.

**Case 2:** $k \geq t + 2$**.** For each $x \in H$, the set $N_F(x) \cap B$ has cardinality at most $t$. Extend it arbitrarily to a $t$-element set

$$A_x \subseteq B, \qquad N_F(x) \cap B \subseteq A_x.$$

This is possible because $k \geq t + 2 > t$.

Choose pairwise distinct real numbers $a_z$, one for each $z \in B$, and associate to each $x \in H$ the polynomial

$$p_x(X) := \prod_{z \in A_x} (X - a_z).$$

When $t = 0$, the product is empty and is interpreted as 1. Each $p_x$ lies in the vector space

$$\mathcal{P}_t := \{p \in \mathbb{R}[X] : \deg p \leq t\},$$

which has dimension $t + 1$. From equation (1) and the assumption of this case,

$$h \geq k \geq t + 2.$$

Consequently the family $\{p_x : x \in H\}$ is linearly dependent.

Choose an inclusion-minimal linearly dependent subfamily, indexed by a set $C \subseteq H$. Put $r := |C|$. There are nonzero coefficients $c_x \in \mathbb{R}$ such that

$$\sum_{x \in C} c_x p_x = 0. \tag{2}$$

Minimality implies that every coefficient $c_x$ is nonzero and that every proper subfamily is linearly independent. Since $\dim \mathcal{P}_t = t + 1$, we also have

$$2 \leq r \leq t + 2. \tag{3}$$

We next extract the combinatorial consequence of equation (2). Suppose that some $z \in B$ belongs to exactly $r - 1$ of the sets $A_x$ with $x \in C$. Evaluating equation (2) at $X = a_z$ annihilates precisely those $r - 1$ terms. The remaining term is nonzero: its coefficient is nonzero, and none of its linear factors vanishes at $a_z$ because the numbers $a_w$ are pairwise distinct. This is a contradiction. Therefore an element of $B$ that occurs in at least $r-1$ of the sets $A_x$ must in fact occur in all $r$ of them.

Define

$$P := \bigcap_{x \in C} A_x, \qquad s := |P|.$$

The preceding paragraph shows that $P$ is exactly the set of elements of $B$ that occur in at least $r-1$ of the sets $A_x$, $x \in C$.

Factor the common polynomial

$$g(X) := \prod_{z \in P} (X - a_z)$$

from the polynomials $p_x$, and write

$$\widetilde{p}_x(X) := \frac{p_x(X)}{g(X)} \qquad (x \in C).$$

The family $\{\widetilde{p}_x : x \in C\}$ is still minimally linearly dependent: any dependence among a proper subfamily would become a dependence among the corresponding original polynomials after multiplication by $g$. Moreover, each $\widetilde{p}_x$ has degree $t-s$. Hence any $r-1$ of these polynomials are linearly independent in the space $\mathcal{P}_{t-s}$, whose dimension is $t-s+1$. It follows that

$$r - 1 \leq t - s + 1,$$

or equivalently

$$s \leq t + 2 - r. \tag{4}$$

By equations (3) and (4), we may choose a set $Y \subseteq B$ satisfying

$$P \subseteq Y, \qquad |Y| = t + 2 - r.$$

Indeed, the required size is at least $s$, and it is at most $t < k$. Now set

$$D := C \cup Y.$$

Because $C \subseteq H$ and $Y \subseteq B$, the union is disjoint, and thus

$$|D| = r + (t + 2 - r) = t + 2 = q.$$

It remains to verify that $D$ is 2-dominating. Let $v \in V(G) \setminus D$. If $v \in H$, then

$$d_F(v, D) \leq d_F(v) \leq t.$$

Now suppose $v \in B$. Since $v \notin D$ and $P \subseteq Y \subseteq D$, we have $v \notin P$. By the characterization of $P$, the vertex $v$ occurs in at most $r-2$ of the sets $A_x$, $x \in C$. The containment $N_F(x) \cap B \subseteq A_x$ therefore gives

$$d_F(v, C) \leq r - 2.$$

Also $d_F(v, Y) \leq |Y|$, so

$$d_F(v, D) \leq d_F(v, C) + d_F(v, Y) \leq (r - 2) + (t + 2 - r) = t.$$

Thus every vertex outside $D$ has at most $t$ nonneighbours among the $q = t + 2$ vertices of $D$. Equivalently, every such vertex has at least $q - t = 2$ neighbours in $D$. Therefore $D$ is a 2-dominating set, and

$$\gamma_2(G) \leq |D| = q = n - m + 1.$$

This proves the theorem. □

*Remark* (Connectedness). The proof uses only the degree sequence of $G$, the complement graph, and a linear-algebraic construction. No step requires $G$ to be connected. Thus the bound holds for arbitrary finite simple graphs, and any connectedness hypothesis in the original formulation is superfluous.